

Understanding Activity Segmentation for Multi-sport Competitions

ABSTRACT

Despite the advances in activity detection, their applications in the sports domain are limited. Athletic environments are fast and challenging. Athletes often perform more than one activity in a single workout, especially if they are training for a multi-sport competition, such as a triathlon. These competitions require an athlete to transition from one activity to another quickly. Current logging applications require the user to select the activity they are about to perform, and start and stop the timer for each activity. This could increase the athlete's transition time, and it would not show the transition duration itself outside of the three activities. This paper explores activity segmentation for multi-sport scenarios. Our goal is to identify the activities and segment a user's workout trace into constituent activities, including the transition periods. We use an Apple Watch to gather inertial sensor data and validate our system in the context of a triathlon. The system was trained and tested on 3 activities (running, biking, and swimming) from 5 different participants. Our system achieves 91% accuracy in detecting the activity, and can accurately identify the start and stop times for each. We also validate our results with data collected from a volunteer at a triathlon.

1. INTRODUCTION

For today's athletes, many resources and wearable devices have been created to assist with training. Numerous motion studies have

covered activity recognition for running, cycling and swimming [1, 2, 3, 4, 5, 6] using accelerometers and gyroscopes, but current applications still depend on the athlete to notify it of what activity they are performing. For most applications, this is sufficient, since athletes tend to do only one activity at a time. An exception to this assumption are multi-sport athletics. We consider the example of a triathlon. Triathlons are a combination of three activities performed back-to-back: running, cycling, and swimming. The order is not fixed, and there is also a transition period between activities which counts against the athlete's overall time. Using an activity tracker which requires the user to define a single activity means triathletes must start and stop each activity as they happen. This could increase an athlete's transition time, and it would not show the transition duration itself outside of the three activities. It has been shown that if athletes are able to accurately divide their activities, and include transition between the activities as part of their training, they can significantly improve their overall time [7].

This paper explores continuous activity segmentation in the context of multi-sport scenarios, using a wearable sensor. Our system identifies the athletic activity a user is performing, based on inertial sensor data from a single limb. We also demonstrate how the workout can be segmented to identify the start

and stop times for each activity, including the transition duration in between. Such a system provides athletes with the ability to train and evaluate performance from a single workout trace.

Related Work. Activity recognition using wearable sensors has been explored previously. In the past, researchers have used accelerometers or gyroscopes to identify or evaluate swimming [6, 8, 9], running [6, 10, 11, 12], or cycling [1, 3, 4, 5], and some success using a wrist mounted sensor [3, 4, 5, 13, 8, 6]. However, they either assume that the activity is known, or focus on daily activities such as sitting, walking, standing etc. Also, the timeliness of these detections wasn’t analyzed. Even recently, researchers have proposed solutions for multiple-activity segmentation in more controlled environments, using rich sensors, such as cameras [14, 15, 16, 17]. Current applications for tracking athletic data such as STRAVA [18] or RunKeeper [19] allow a user to track different activities but not all in the same session. In the case of STRAVA, a user can manually divide a workout after the fact, but they would need to keep track of start and stop times for each activity, as well as the duration for which they were changing equipment or taking a break.

Challenges. Determining transitions between sequential activities is challenging compared to simply detecting an activity. In a multi-sport scenario, the transitions between activities are not well defined, and an athlete could practically do anything in-between activities, including stretching, jogging in-place etc. On the other hand, even simple actions such as drinking water, walking, and tying their shoes can occur during an athletic activity, such as running. A major challenge is distinguishing when these actions appear in-between activities, and when they mark a change in activity. Moreover, for an athlete, knowing the exact start and stop times for each activity (including transitions) is crucial, and must be determined accurately. Transi-

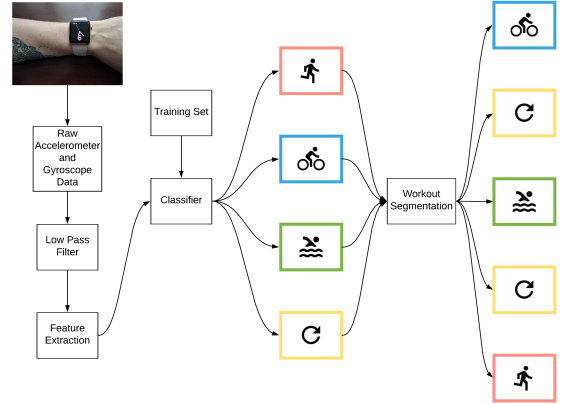


Figure 1: System Overview: A single workout is segmented into constituent activities, including the transitions between them.

tion times count against the overall time, and lack of techniques to measure these times offers serious disadvantages to athletes training for such events. Our proposed system addresses these challenges by not only identifying the activities in the context of a triathlon, but also determining the transitions between activities. Predicted labels from a classifier are often computed independently for small durations, and can result in stray misclassifications. Therefore, we process classifier outputs to discard misclassifications caused by transition-like actions within an activity.

Applications. Other multi-sport competitions, such as duathlon, swimrun, and rowathlon can also benefit from workout segmentation. Endurance races, for example, Spartan Race or Mud Run are also some of many potential applications of such a system. Additionally, the techniques proposed here can also be used in team sports, to monitor the total amount of time a player was running, walking, or sitting on the bench.

Our goal is to allow a user to capture one workout trace with multiple activities, which is then segmented into individual activities automatically, as opposed to manual definitions. In summary, the contributions of this

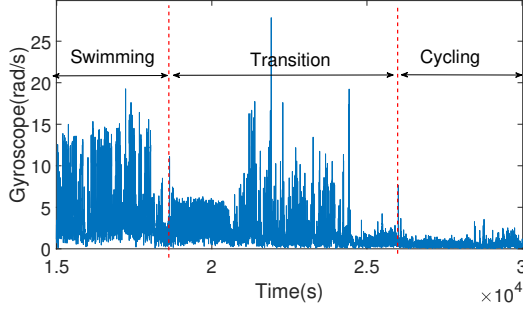


Figure 2: The magnitude of the Gyroscope data while transitioning from one activity (swimming) to a second activity (cycling).

work are:

- Segmenting a single workout trace by detecting and classifying 3 athletic activities and the transition between them.
- Evaluating the system on real world multi-sport workout data.

2. DESIGN AND METHODOLOGY

Figure 1 shows an overview of our system. We train a classifier to detect three activities and the transitions between them as a fourth class. The outputs from the classifier are used to segment the workout trace, and determine the duration of the activities and the transitions. In this section, we discuss the details of our algorithm.

2.1 Data Preprocessing

Leveraging Swift 4.0 and HealthKit, we designed an application that collects accelerometer and gyroscope data from the Apple Watch 3 continuously. HealthKit workout keeps all background processes alive as it needs to keep collecting data even when the screen is off. Our WatchOS application offers a button to start and stop the workout. There is also a timer to show how long it has been since the workout started. The raw accelerometer and gyroscope readings along all three axes were

captured at $30Hz$, and logged into a file on the watch. The data collection end of our system was tested thoroughly with sessions of long duration to ensure the data would be saved. We used a lowess regression filter to smooth the accelerometer and gyroscope data along all three axes. We then computed the magnitude for accelerometer and gyroscope using the filtered readings. The gyroscope magnitude is given by: $\mathcal{G} = \sqrt{G_x^2 + G_y^2 + G_z^2}$.

Figure 2 shows a small trace of gyroscope data, with swimming, transition, and cycling labeled.

2.2 Feature Selection

In order to classify the user’s activity, we used a sliding window technique [1, 3, 4, 20, 12, 13, 10, 8, 6] and extracted features from each window. We tested with window sizes of 3 seconds, 5 seconds, and 10 seconds. The window was shifted by 1 second at a time. From each window we compute temporal and frequency-based features from both the accelerometer magnitude and the gyroscope magnitude. For each window these features were: mean distance between peaks, deviation of distance between peaks, deviation of peaks amplitude, mean of peaks amplitude, root-mean-square level of the window, watch orientation computed from accelerometer readings, and the maximum power spectral density. Figure 3 shows three features for all four classes in a triathlon. We can see that the mean peak distance is most distinctive for swim, but exhibits a much larger range for transitions.

2.3 Activity Segmentation

To classify each window, we use supervised learning. In the context of a triathlon, we define four classes: cycling, running, swimming, and transition. Transitions are referred to as time intervals right after an activity has been completed and before the next one is started. We divide our data into training and test subsets. The feature vectors are computed from the training set in 5 second and 10 second

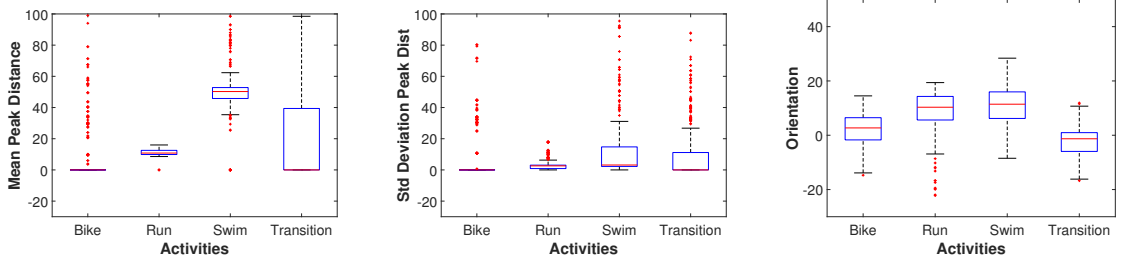


Figure 3: 11 features were extracted from the accelerometer and gyroscope data. The mean distance between peaks, the standard deviation of the distance between peaks for the accelerometer, and watch orientation showed the greatest differences across two or more activities.

windows. For our classifier, we use bootstrap aggregating, also known as bagging. The classifier was chosen based on empirical performance. It is an ensemble classifier that melds several weak learners into one high-quality learner. The classifier is constructed using five fold cross validation on the training set. The training set was collected by capturing different activities from volunteers. We obtain 98% accuracy on our training set for the 10 second window and 95% accuracy for the 5 second window.

We then divide our data set into similar sized windows and extract features for each window. Each window is classified by our classifier described earlier. We perform two-stage post processing of the predicted labels. In the first stage, we analyze the probability scores returned by the classifier. If the score for the predicted class is lower than 60%, we split the window into two and classify the first sub-window to obtain the predicted label for that window. These predictions are then used for determining the start and stop times of each activity. Since these activities are performed sequentially, we process them with respect to predicted classes in the windows before the current window. Transition-like activities, such as drinking water or walking, are common when a user is performing another activity and thus must be discarded

when they appear between two running segments. We make this decision based on the duration of this transition window. A transition prediction must appear multiple times in quick succession before it is determined that a transition may have begun. If the preceding activity is again detected with high confidence over a short period, it is considered to be a part of that same activity and the short period of transition-like activity is discarded. If a transition lasts longer without sustained re-occurrences of the previous activity, it is considered to mark a switch in activities. Since the accuracy of the transition classification is so low, we also viewed spurious appearances of activities other than the current activity as possible transition events. The predicted labels are processed using these practical constraints to output the start and stop times of each activity. The proposed activity segmentation is performed offline.

3. EXPERIMENTAL SETUP

We collected data from four volunteers for running and cycling, three male and one female. All volunteers have trained for and completed a triathlon in the past. The data we collected consisted of the x, y and z axes from both the accelerometer and gyroscope on the Apple Watch, at a rate of 30Hz. Each session was recorded using a GoPro Hero 4

		Accuracy: 90.95%			
Output Class	Transition	64.4% 826	6.9% 374	1.6% 68	3.7% 16
	Bike	19.0% 243	92.9% 5010	1.5% 64	0.0% 0
	Run	9.2% 118	0.1% 8	96.0% 4182	1.6% 7
	Swim	7.4% 95	0.1% 3	0.9% 41	94.7% 409
		Transition	Bike	Run	Swim
		Target Class			

Figure 4: Classifier accuracy on unseen test data ~91%. The test data included 5 different traces that combined two or more activities.

Silver at 1080p - 60 frames per second. These recordings were used to establish ground truth. We manually labeled the exact times when the user pushed the start button on the watch and their subsequent activities and transitions. Each volunteer completed a combined workout, featuring two or more activities, in any order. Each activity was approximately 10 minutes long, with a short transition period between activities, for a maximum of 25 minutes. During the transition periods, volunteers performed actions such as changing and tying their shoes, stretching, walking around to catch their breath, sitting down, and drinking water. Collecting swimming data was challenging due to the lower temperatures and availability of volunteers. Swimming data was collected from 2 volunteers, both male. Data collection was done in a semi-Olympic swimming pool. Overall, we collected a total of 45 minutes of swimming data. We also collected data from one volunteer at a triathlon, used as test data for our evaluation.

4. EVALUATION

In order to evaluate our system, we cap-

tured a real triathlon for one of our volunteers. We used this triathlon data in addition to other data collected by our volunteers, as described in the earlier section. We selected five traces to constitute our test set. All the results presented here are derived from the 5 trace test data set. In this section we discuss the performance of our system.

Activity Classification. We measured the accuracy of our system in identifying the classes. The results are shown in the form of a confusion matrix in Figure 4. The accuracy was measured by comparing the ground truth labels and predicted labels for corresponding time windows in the test trace. The overall accuracy of our system was ~91%. We can see that core activities - running, swimming, and cycling, are identified more accurately than transitions. Transitions are hard to define, since many actions performed during a transition can appear elsewhere in the trace too. Another factor that affects our results is how we define the start and stop of each event. According to our labeling convention, cycling only starts when the athletes start riding their bicycles. However, the time duration when the athlete is walking their bicycle to the start line is also classified as cycling by the algorithm, while it’s defined as transition in our ground truth labels. The similarity between these traces can be seen in Figure 5. The accuracy numbers reported here will be affected by how the temporal boundaries between classes are specified. The official times for triathlons are similarly blurred. When an athlete leaves the water, the point at which the end of the swim time is recorded could be a dozen yards away. The only consistent measurement is that the end of each activity happens somewhere between the course and the transition area.

Detection Delay. In addition to classifying an activity, athletes are also interested in learning the time they’ve spent performing each activity, and the time spent transitioning from one activity to another. We calculate the

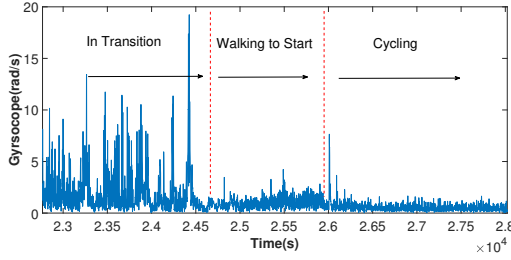


Figure 5: Walking with the bike vs riding the bike. The traces show a great deal of similarity due to the position and stability of the wrist while the athlete holds the handlebars.

detected start and stop times for each activity. The mean start and stop times for each activity are shown in Figure 6. Negative values denote that the activity was detected earlier than it started, while positive delay represents that an activity was detected by our system after it started/stopped.

The start of the bicycling activity was detected earlier for the same reason mentioned in the previous subsection. Athletes must walk their bike from the transition area to the start line before they start riding. Since their hands are in the same position as if they were riding the bike, the algorithm classifies this as a bike activity due to the similarity in trace (shown in Figure 5). On the other hand, our manual labels only classify an activity as biking if the user was riding it. The stop times of biking activity are detected later because of the same reason; athletes walk their bikes after they cross the dismount line.

Similarly, for swimming, the finish line is often a little far. Athletes have to walk/run a certain distance to the finish line after they get out of the water. We detect an early stop to the swimming event because while the person did stop swimming, the time they spend getting to the finish line is still counted against their swimming times. On a similar note, the start line for swim can be a distance away from the water, causing the actual swimming

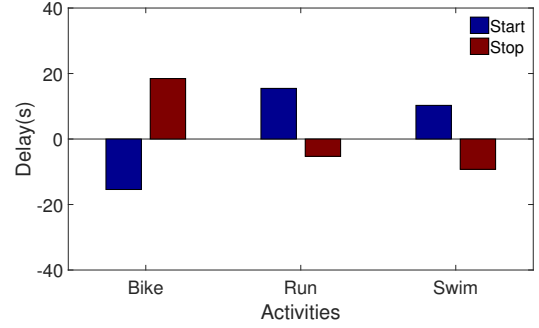


Figure 6: The recognition delay on test data. Negative values denote early detection by our system, while positive values represent a delayed detection.

activity to happen several seconds after crossing the start line. The ground truth labels were assigned based on when an activity started and stopped, and not when the start and finish lines were crossed. This is because while training, the start and finish lines are not as well defined as during an actual race. With precise timings for each activity, trainees may benefit from knowing the exact time spent performing each of them.

5. DISCUSSION AND CONCLUSION

We proposed a system that processes time-series data for a multi-sport scenario to identify the constituent activities, and determine the duration of each activity. Our classifier achieves greater than 90% accuracy for distinguishing between swimming, biking, running, and transitions. We include transitions in our analysis because athletes who are training for such competitions are interested in knowing the time spent performing each activity and the transitions from one to another.

Sports scenarios are challenging due to their dynamic nature and less defined temporal boundaries between events. Our system can accurately detect the start and stop times for each activity. Due to differences in labeling convention, we observed some discrepancies, such

as, a time-window being classified as cycling if the volunteer's hand is on bicycle when he is walking the bike or holding the left arm up to look at the watch while running is classified as transition. These times may not match up to the official triathlon timings. This is because official timings are measured from start line to finish line, while our system detects start and stop for the activity. We believe that computing the time spent performing each activity is more useful for trainees looking to measure or improve their performance.

There are several ways of addressing the discrepancy in timings and improving the system accuracy. Since limbs are more involved in transitional activities, wearing an inertial sensor on the ankle instead of the wrist may perform better. For example, it might be able to distinguish between walking a bicycle and riding it. We intend to explore this alternate sensor mounting position in the future.

We found that even after collecting 2 hours of data via our app, the Apple Watch was able to keep running for the rest of the day without requiring a recharge. This implies that our data collection system is not harsh on the battery of the watch. We will perform a quantitative analysis of power consumption in the near future.

This paper explores the possibility of applying activity detection to sports. By automatically segmenting a workout trace, it can measure the fraction of time spent in each activity as well as the transition periods. Such analysis has been found useful in the past [7]. We believe that such a system has immense potential in athletic training and performance evaluation.

6. REFERENCES

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